

Unit 4

CHEMICAL ROCKET PROPULSION

A rocket is a vehicle which works on the principle of Newton's third law which states that "for every action there is an equal and opposite reaction". A rocket is a sophisticated jet powered vehicle intended to carry equipments or crew beyond earth and into the space to perform a specific application or task. Space is generally defined as starting at about 100km (60 miles) above Earth (an arbitrary point sometimes called the Karman line), which is where conventional planes would struggle to make enough lift to stay in the air. Space is the place effectively beyond Earth's reach, beyond most of its gravity and atmosphere.

A rocket engine in its simplest form consists of a propellant system, combustion chamber and a nozzle. Here the fuel and the oxidant are thoroughly mixed in the combustion chamber which when ignited results in large volume of exhaust gases that passes through the nozzle, thus creating thrust. The force responsible for pushing or propelling the vehicle forward is called as the Thrust of the engine. Common applications of rockets are Fireworks, Weaponry, Aircraft seat ejection systems, Satellites for space exploration, planetary and interplanetary missions.

CLASSIFICATION OF ROCKET PROPULSION

1) BASED ON ENERGY SOURCE EMPLOYED

- a) **CHEMICAL ENERGY**
 - » Solid Propellant Rockets
 - » Liquid Propulsion System
 - » Gaseous Propulsion System
 - » Hybrid Propulsion System
- b) **NUCLEAR ENERGY**
 - » Nuclear Fusion Propulsion System
 - » Nuclear Fission Propulsion System
 - » Radioactive Isotope Decay
- c) **SOLAR ENERGY**
- d) **ELECTRIC ENERGY**
 - » Electric Thermal Propulsion System
 - » Electro-Static Propulsion System
 - » Electro-Magnetic Propulsion System

2) BASED ON THE METHOD OF PRODUCING THRUST

- a) **THERMO-DYNAMIC ROCKET**
 - » Chemical Rocket
 - » Nuclear Rockets
 - » Solar Rockets
 - » Electro-Thermal Rockets
- b) **ELECTRO-DYNAMIC ROCKETS**
 - » **Electro-Static Rockets**
 - Ion Thrusters
 - Hall Thrusters
 - » **Electro-Magnetic Rockets**
 - Magneto-Dynamic Rocket Propulsion System
 - Pulsed Propulsion Thruster system

3) BASED ON THE PHYSICAL STATE OF THE PROPELLANTS

CHEMICAL ROCKETS

- **SOLID ROCKETS**
 - **Homogenous Propellants**
 - Single Base
 - Double Base
 - **Heterogeneous Propellants**

- **LIQUID PROPELLANT ROCKETS**
 - **Monopropellant**
 - **Bi-Propellants**
 - Storable Propellants
 - Cryogenic Propellants
 - Hypergolic Propellants
- **GASEOUS ROCKETS**
 - Cold Gas Rockets
 - Warm Gas Rockets

4) BASED ON THE MISSION / APPLICATION

a) Military Rockets

- **Reconnaissance Rockets**
- **Missiles**
 - » **Strategic Missile**
 - Ballistic Missiles
 - IRBMs
 - ICBMs
 - » **Tactical Missiles**
 - » **Crew Ejection Capsule/Seat Ejection in Military aircrafts**
 - » **Ejection of stores and armaments**
 - » **Under Water rockets for Torpedoes and Missiles**

b) Non-Military or Governmental Rocket/Satellites

- Weather Observation satellites
- Geopositioning satellites
- Sounding rockets
- Decoy rockets
- Spin rockets
- Vernier rockets

c) Space Exploration Rockets

- Earth Satellites
- Lunar Satellites
- Inter-planetary Satellites
- Trans-Solar Satellites

d) Commercial Satellites

- Communication Satellites

5) BASED ON FUNCTION

a) Primary Functions

- Orbital Insertion
- Orbit Manoeuvring
- Boosting

b) Secondary Functions

- Attitude Control
- Spin Control
- Momentum wheel and Gyro unloading
- Stage Separation
- Trajectory Transfer
- Orbit Station Keeping

6) BASED ON SIZE AND RANGE

- a. Small Size and Short Range Rockets
- b. Large Size and Long Range Rockets

7) BASED ON NUMBER OF STAGES

a. Single Stage Rockets

b. Multi Stage Rockets

- Two-Stage Rockets
- Three-Stage Rockets
- Four-Stage Rockets
- Five-Stage Rockets

Types of Propulsion Systems Based on the Source of Energy

1. Solid Propellant Rockets
2. Liquid Propellant Rockets
3. Gaseous Propellant Rockets
4. Hybrid Propellant Rockets
5. Nuclear Propellant Rockets
6. Electric Propellant Rockets
7. Solar Propellant Rockets

Solid Propellant Rockets

The technology of solid rocket propulsion is well understood and disseminated that many companies or government arsenals are now capable of designing developing, and manufacturing solid rockets in several categories. Solid propellant rocket motors are being built in approximately 35 different countries today, compared to only three countries about 50 years ago. A category of rockets which use propellants in solid form or solid state are known as solid propellant rockets. In rocket industry, solid propellant rockets are usually termed as Motors.

In solid propellant rockets, the propellants namely, the fuel and the oxidizer which are in the solid form are thoroughly grinded and reduced to fine granules. The solid propellant charge in the form of fine granules are called as Grains and it contains all the chemical elements for complete burning. The Grain is thoroughly packed and stored in the combustion chamber and sometimes hermetically sealed in the chamber for long-time storage up to 20 years. The hardened solid grains stored in the combustion chamber case typically accounts for 90% of the total motor mass. A cylindrical star shaped cavity is provided for combustion in the centre along the axis of the shell. Different shapes and geometric features like slots, grooves, holes etc can be provided in the cavities so that various burning time profiles can be obtained which alters the thrust profile suitable for a given mission. The inner surfaces of the case (really a pressure vessel), which are exposed directly to hot gas, have a thermal protection or insulation layer to keep the case from becoming too hot, in which case it could no longer carry its pressure and other loads. The case is either made of metal (such as steel, aluminium or titanium) or a composite fiber-reinforced plastic material. An electrically activated igniter located at the top provides the energy to start the combustion process. Initial burning takes place at the internal surfaces of the cylindrical perforation of the propellant.

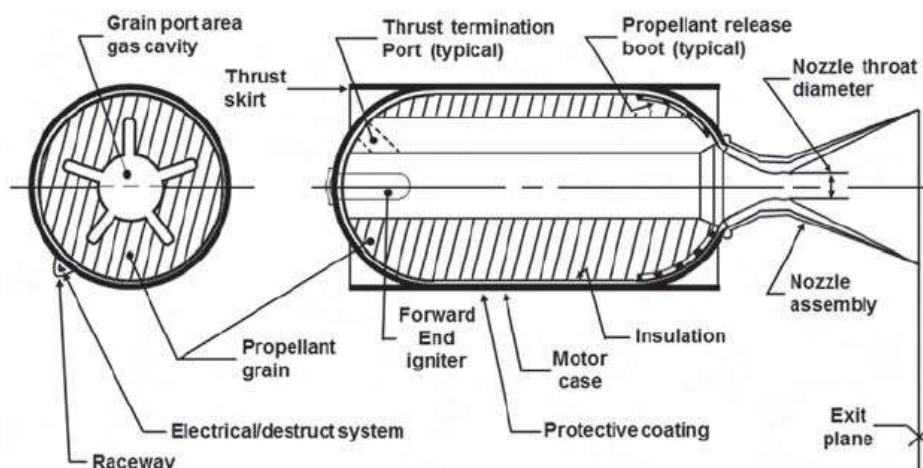


Fig : Solid Propellant Rocket

Once ignited, it usually burns smoothly at a predetermined rate on all the exposed internal surfaces of the grain. The motor combustion proceeds in an orderly manner until essentially all the propellant has been consumed. The energy from a high-pressure combustion reaction of propellant chemicals permits the heating of reaction product gases to very high temperatures (2500 to 4100°C). The hot reaction gases flow along the perforation or port cavity toward the nozzle. These gases subsequently are expanded and accelerated to very high velocities (1800 to 4300 m/sec) through supersonic nozzle. The nozzle is usually made of high temperature materials (usually a graphite and/or an ablative material to absorb the heat) to withstand the high temperatures and the erosion. The majority of all solid rockets have a simple fixed nozzle, as shown here, but some nozzles have provision to rotate it slightly so as to control the direction of the thrust to allow vehicle steering. Motors come in many different types and sizes, varying in thrust from about 2 N to over 4 million N). The solid motor is used mainly as a booster for launch vehicles. Solid motors are almost never used in space because they are not controllable. The boosters are lit and then they fire until all the propellant has burned. Their main benefits are simplicity, a shelf life which can extend to years as in the case of missiles, and high reliability.

Advantages

1. In comparison to liquid rockets, solid rockets are usually relatively simple, are easy to apply (they often constitute most of the vehicle structure), and
2. Higher thrust at relatively low cost
3. Free from moving parts- no feed systems or valves
4. Comparatively lighter for shorter range and small size
5. On account of high density much larger quantity of propellant can be packed into a small space.
6. Specific gravity is as high as 2
7. Requires Less servicing
8. Vibrations arising from turbo-pump and liquid flow are absent
9. Problems arising from sudden emptying of propellant tanks are absent

Disadvantages

1. Specific impulse is comparatively lower than Liquid Rockets
2. Exhaust velocities are less, so less performance
3. Impossible to turn-off the motor midway, once started engine cannot be stopped
4. Minor malfunctioning or accident will abort the whole mission
5. Difficult to control combustion and regulate thrust
6. Not re-usable
7. Only for small size and short range rockets
8. Long range rockets are very heavy
9. Transporting and handling is very difficult
10. Refuelling not possible
11. Nozzle suffers from erosion of grain particles

Properties of Solid Propellants

1. Propellant materials should be abundantly available
2. It should be easy and safe to handle the propellants
3. It should be easy and safe to grind, mix, heat, trim, pack and store the propellants. Also it should be practical to cast or extrude the solid propellants in desired shapes and sizes.
4. Physical and chemical properties of the propellant mixture should not change considerably during processing
5. Heat of chemical reaction between the fuel and oxidizer should be very high so that the high combustion temperatures and specific impulse are obtained.
6. Properties of the solid propellants should not deteriorate with time.
7. They should be chemically inert before ignition and should not ignite with slight impact, shock or pressure.
8. They should possess high mechanical strength.
9. They must have comparatively lower molecular weight and higher density which yields high values of specific impulse and also result in lighter structure.
10. In case of military applications, the exhaust gases should be smokeless and colourless.
11. They should not be hygroscopic i.e., they should not react with atmospheric air and moisture.

12. There should not be any chemical reaction between the constituents of the propellant materials while in storage.

Applications of solid propellant rockets

Application of Solid Rockets	
Large booster	Space launch vehicles;
Second stage Motors	Lower Stages of long-range ballistic Missiles
High-altitude Motors	Upper stages of multistage Ballistic missiles, space launch Vehicles; space manoeuvres
Tactical missiles	High acceleration: short-range bombardment, antitank missile Modest acceleration: air-to-surface, Surface-to-air, short range Guided surface-to-surface, air-to-air missiles
Ballistic missile defence	Long- and medium-range ballistic missiles
Gas generator	Pilot emergency escape; Push missiles from submarine launch Tubes or land mobile canisters; Automotive air bags

Solid Propellant Types and Ingredients

Gun Powder was the first propellant that was used as both an explosive as well as propellants for Solid Propellant rockets. Gun Powder is a mixture of Salpeter, which is nothing but Potassium Nitrate (potassium, nitrogen, and oxygen), Charcoal and sulphur. When 75% potassium nitrate is mixed appropriately with 15% carbon (charcoal) and 10% sulfur, the resultant black powder has explosive properties. Gun Powder was first used in the preparation of explosive grenades, bomb and ultimately ended up as a propellant for rockets since they were able to accelerate projectiles from the guns and cannons. Due to this gun powder ended up as propellants for accelerating missiles and rockets. However, this old type of propulsion had serious disadvantages

1. Highly Unpredictable- Developed through Trial and Error, No history on Burning Characteristics, Stability
2. Developed extremely dirty gases
3. Acidic residue deposits inside the barrel caused erosion of the inner surfaces
4. Caused considerable fouling of gun barrels
5. Emitted high muzzle flash-Disclosed the location of the gun to the enemy

In 1887, Alfred Nobel formulated a new type of propellant based on the nitrocellulose and nitroglycerine, which led to the development of new propellant formulations.

Basically, solid propellants can be classified as

1. **Homogenous mixtures or organic substances-**
 - a. Single base propellants
 - b. Double base propellants
 - c. Triple base propellants
2. **Heterogeneous or Composite propellants**
 - a. Composite propellants

Single Base Propellants

Compounds that are developed from one single base are called as Single Base Propellants. Single Base Propellants are mainly comprised of Nitrocellulose (NC) or more exactly nitrates of cellulose. When cellulose is exposed to nitric acid or any other powerful nitrating agents, a highly flammable compound, nitrocellulose is formed. This nitrocellulose compound with nitrogen content varying between 12-15% is gelatinized by mixing in ether alcohol mixture. Later, the gelatinized Nitrocellulose is usually combined with a 10-12% of plasticizer like Dibutyl Phthalate/ Dibutyl Sebacate/Camphor etc, plasticizer mainly acts as a binder and helps bind the whole mixture together. This compound of Nitrocellulose and a plasticizer is mixed with

diphenylamine, which turns the whole compound into dough. Further, the gelatinised dough is extruded through dies of required size and shape and then chopped and dried. To improve the burning characteristics of double and triple base propellants, Tar or rubber was used as fuel that was combined with an oxidiser.

The explosive nature of the SBP is dictated by the amount of Nitrogen content in the compound. The higher the nitrogen content, the higher the heat of explosion. More nitrogen is present in the molecule and more oxygen can be used to oxidize the reducing agents. Based on the Nitrogen Content, there are two types

- a. **Grade A .pyrocellulose.: nitrogen content between 12.3 and 12.8%**
- b. **Grade B .guncotton.: nitrogen content higher than 13.35%.**

Advantages of NC

- a. No smoke,
- b. Little muzzle flash,
- c. More energy,
- d. Reliability in use,
- e. Long storage life.

Applications

- a. Used in rifles and guns
- b. In nearly all calibres of handgun, rifle, machine gun, aircraft- and antiaircraft gun, cannons and howitzers

Double Base Propellants

By adding nitrates of glycerine or just Nitro-glycerine to the Nitrocellulose, a double base propellant can be obtained. Such propellants consist of two bases, namely Nitro-glycerine and Nitrocellulose, and hence the name. Double base propellants are obtained by kneading NC and NG with some stabilisers, ballistic modifiers etc, and then it will be carpet rolled at elevated temperatures. Finally, the mix is extruded through dies. Also, the double base propellants can be produced through casting techniques. DBP show low elongation and high modulus especially at lower temperature and such grains are used in the free-standing configuration. For bigger and larger application, Cast DBPs are used and for smaller application, extruded DBPs are used.

The heat of explosion for this reaction is about 6100 J/g. This figure is 1.5 times higher than that of NC. This gain of energy is positive and fulfils the military demand for increased bullet velocity and prolonged range of shooting.

Applications

- 1. Large calibre cannons and howitzers- shape of tubes or slotted tubes
- 2. Pistols and grenade launchers- flake powders are found in ammunition
- 3. Rockets and missiles: the antitank missile- rolled up db propellant sheets in its numerous small motors
- 4. The ejection of the missile from the launcher is done by a charge of short tubes of db propellant
- 5. Rockets and missiles: the antitank missile-large sized tubes

Triple base propellants

To make the propellants more energetic, a third energetic ingredient, namely nitroguanidine (NQ) also known as picrate is added to the double base propellant consisting of NC and NG. About 40% of weight of the propellant, NG is added.

Advantages

- 1. Exhibit good mechanical properties
- 2. Good ballistic stability

Though, SBP and DBP exhibit good burning characteristics, their size and geometry is limited, that limits the opportunity for large scale applications like the solid rocket boosters used as launch vehicles, primary stage of

a rocket or for a long range missile. Further, DB propellants are affected by brittleness, cracking, and high pressure index of burning, that will be fatal for the rocket performance. Due to these demerits, homogenous propellants are not preferred for large scale applications.

A new type of propellant namely, the heterogeneous propellants have been developed especially for large scale applications. Like any other propellant, even a heterogeneous propellant will consist of the basic fuel and oxidiser. Generally, nitrates and perchlorates (NH_4NO_3 , KNO_3 , KClO_4 , NH_4ClO_4) are used as oxidisers and polymers, plastics & PVC are used as fuels. But apart from the oxidiser and fuel, it will also consist of host of other additives like binders, inhibitors, opacifiers, plasticizers, metallic materials, burn rate modifiers, curatives, phase changers, solvents etc that will improve the overall burning characteristics, manufacturability and versatility of the propellant.

The various constituents of a heterogeneous or composite propellant are detailed as follows:

Oxidiser

- The oxidiser is the major constituent of composite solid propellant system. Oxidiser accounts for 68 to 70% by weight. Hence, it is one of the important constituent of any propellant mixture.
- Primarily, the requirement of oxidiser is that it should have high oxygen content and easily decompose as and when required to produce necessary oxidising elements to maximise the energy release from the reaction.
- The two main oxidizing compounds used in composite propellant manufacture are ammonium perchlorate (AP- NH_4ClO_4) and ammonium nitrate (AN). The most frequently used is AP because it is more stable and safer to handle.
- One can see that during the burning process a lot of hydrochloric acid (HCl) is developed. With the moisture in the air, this leads to an intense white smoke, so the trajectory and the starting point of the missile can easily be observed. Adding HCl-suppressant substances, like magnesium-aluminum alloys or sodium nitrate, the formation of this white smoke will be reduced. This problem is very important for tactical missiles, but is less important for space exploration. The heat of explosion of AP is 1115 J/g, which is less than that of NC or NG. Because composite propellant must burn for a relatively long time, the burning rate has to be low and a high heat of explosion is not at the forefront.
- AN on the other hand, does not show the problem of hydrochloric acid production, but its main drawback are the different crystal modifications. A crystal modification change in a rocket motor is fatal. It will influence the physical properties, cracks in the propellant could occur enlarging the burning surface. Under operational conditions the propellant produces more gases than the nozzle can manage and the rocket or missile finally explodes. The use of AN for rocket motor technology also needs the addition of a crystal structure stabilizer, *e.g.* potassium nitrate •
- Oxidisers that are added in the propellant may come in various mesh sizes. The particle sizes of the oxidisers are mainly varied to control the combustion rate of the propellant. Oxidisers having fine mesh size (rapidly combine) will increase the rate of combustion and vice versa.

Fuels

- Next important constituent of the propellant mixture is the fuel. Fuel constitutes of 14-20% of the propellant by weight. Powdered spherical aluminum is the most common fuel used. It consists of small spherical particles (5 to 60 μm diameter) and is used in a wide variety of composite propellant preparation.
- During rocket combustion this fuel is oxidized into aluminum oxide. These oxide particles tend to agglomerate and form larger particles. The aluminum increases the heat of combustion, the propellant density, the combustion temperature, and thus the specific impulse.
- Boron is a high-energy fuel that is lighter than aluminum and has a high melting point (2304°C). Due to the high melting point, its difficult to burn the boron fuel, but with particle size being very small, it can be achieved.

- Beryllium burns much more easily than boron and improves the specific impulse of a solid propellant motor, usually by about 15 sec, but it and its oxide are highly toxic powders absorbed by animals and humans when inhaled.

Binders

- After, fuel and oxidizer, the next important constituent is the binder. The binder is an essential ingredient of composite solid propellants. Nearly 10 to 15% of the composite solid propellant is comprised of binders.
- Binder is actually in the form of glue or resin. Soon when all the propellant constituents are added, it will help in binding all these particles and keep them intact.
- After they are mixed with the solid ingredients, cast and cured, they form a hard rubber-like material that constitutes the grain.
- Polyethers, polyesters and poly-butadiene have been used as binders. Polyvinylchloride (PVC) and polyurethane (PU) were used 40 years ago and are still used in a few motors, mostly of old design.
- Binders also act as fuel in the propellant. Besides, they help in improving the reliability, mechanical properties, propellant processing complexity, storability, aging, and costs.
- HTPB has been the favorite binder in recent years. Some polymers undergo complex chemical reactions, crosslinking, and branch chaining during curing of the propellant.
- Polymerization occurs when the binder monomer and its crosslinking agent react (beginning in the mixing process) to form long-chain and complex three-dimensional polymers.

Plasticisers

- A plasticizer is usually a relatively low-viscosity liquid organic ingredient which is also a fuel.
- Plasticisers are mainly added to improve the processability of the dense slurry of the propellant mixture. Since, a lot of different additives are added to the mixture, it will make the propellant slurry very strong and reduce the fluidity of the mixture. To make it more fluid and to improve the mixing between the constituents, plasticisers are added.
- Plasticisers are high boiling, non-volatile, low viscous, low molecular weight substances. The modification of the flow characteristics of the propellant slurry is very important for realising defect free propellant grains.
- Binders such as polymers, polybutadiene are used to hold the fuel and oxidizer powders together.

Curatives

- The percentage of curatives added is insignificant in comparison with other constituents. The percentage of curatives will be between 0.2 to 3% of the weight. Though they are small in percentage, they have a greater role to play that affects the physical properties, manufacturability and aging characteristics of the propellant.
- Adding a curative or a curing agent causes the prepolymers to form longer chains of larger molecular mass and helps in providing interlocks between chains.
- The curatives react with the functional groups of the binder and forms cross-linked polymeric network structure. The number of crosslinks in the network decides the mechanical behaviour of the polymer. Propellant formulations utilise a crosslinking agent, which reacts with the curatives and helps to form three dimensional network structures.

Metallic fuels

- Metallic fuels improve the energetics or the heat of combustion of the reaction. In doing so, the maximum temperature is increased, that increases the chemical energy released during the exothermic reaction.
- Increasing the combustion temperature will ultimately increase the specific impulse of the propellant.
- Aluminium and Beryllium are the most widely used metallic additives in composite solid propellant formulations.
- Aluminium content of 18 to 20% in the propellant is found to improve the specific impulse by 20%.
- Addition of these metals also tends to increase the grain density which decreases the volume required for a given value of the total impulse.
- Presence of aluminium oxide particles helps to attenuate the undesirable acoustic oscillations of the combustion products and thereby eliminate combustion instability. Other important metallic fuel

choices include magnesium, boron and beryllium. Metallic wires of aluminium, silver and copper are also used to enhance burn rate of the propellant.

Burn rate modifier

- Burn rate modifier is another important constituent, added to modify the ballistic properties of the propellant.
- Ferric oxide, ferric acetyl acetonate, ferrocene and copper chromite are usually used in propellant formulations for this purpose.
- A burning-rate catalyst or burning-rate modifier helps to accelerate or decelerate the combustion at the burning surface and increases or decreases the value of the propellant burning rate. It permits the tailoring of the burning rate to fit a specific grain design and thrust-time curve.
- Some, like iron oxide or lead stearate, increase the burning rate; however, others, like lithium fluoride, will reduce the burning rate of some composite propellants. The inorganic catalysts do not contribute to the combustion energy, but consume energy when they are heated to the combustion temperature.

Anti-oxidants

- Anti-oxidants are added to propellant formulations to improve the ageing characteristics by preventing oxidative degradation. Phenyl- β -naphthylamine, paracresol are some of the anti-oxidants used.
- Inhibitors or restrictors are used to prevent or regulate the burning of propellant grain at some stipulated surfaces.
- An opacifier may be added to absorb heat that may otherwise be transmitted through a translucent grain resulting in unpredictable burning.- prevent radiation heating at places other than the burning surface
- Bonding agents are additives to enhance adhesion between the solid ingredients (AP or A1) and the binder.
- Stabilizers are intended to minimize the slow chemical or physical reactions that can occur in propellants.
- Catalysts are sometimes added to the cross linker or curing agent to slow down the curing rate.

Commonly used propellant additives

Type	Percent	Acronym	Typical Chemicals	Curing agent or crosslinker, which reacts with polymer binder		MAPO	Methyl aziridinyl phosphine oxide
Oxidizer (crystalline)	0-70	AP	Ammonium perchlorate	0.2-3.5		IPDI	Isophorone diisocyanate
		AN	Ammonium nitrate			TDI	Toluene-2,4-diisocyanate
		KP	Potassium perchlorate			HMDI	Hexamethylene diisocyanide
		KN	Potassium nitrate			DDI	Dimethyl diisocyanate
		ADN	Ammonium dinitramine			TMP	Trimethylol propane
Metal fuel (also acts as a combustion stabilizer)	0-30	Al	Aluminum	Burn-rate modifier	0.2-3	BITA	Trimesoyl-(2-ethyl)-aziridine
		Be	Beryllium (experimental propellant only)			FeO	Ferric oxide
		Zr	Zirconium (also acts as burn-rate modifier)			nBF	n-Butyl ferrocene
Fuel/Binder, polybutadiene type	5-18	HTPB	Hydroxyl-terminated polybutadiene	Explosive filler (solid)	0-40		Oxides of Cu, Pb, Zr, Fe
		CTPB	Carboxyl-terminated polybutadiene				Alkaline earth carbonates
		PBAN	Polybutadiene acrylonitrile acrylic acid			HMX	Alkaline earth sulfates
		PBAA	Polybutadiene acrylic acid			RDX	Metallo-organic compounds
Fuel/Binder, polyether and polyester type	0-15	PEG	Polyethylene glycol	Plasticizer/Pot life control (organic liquid)	0-7	NQ	Cyclotetramethylenetetranitramine
		PCP	Polycaprolactone polyol				Cyclotrimethylenetrinitramine
		PGA	Polyglycol adipate			DOP	Nitroguanadine
		PPG	Polypropylene glycol			DOA	Diethyl phthalate
		HTPE	Hydroxyl-terminated polyethylene			DOS	Diethyl sebacate
		PU	Polyurethane polyester or polyether			DMP	Dimethyl phthalate
						IDP	Isododecyl pelargonate

Propellant Burning Rate r

The thrust required by a rocket is not constant rather varies throughout its mission. For some missions, the thrust required by the rocket will be high during the launch. As the rockets altitude increase the thrust decreases due to the decrease in the gravity, decrease in the weight of the rocket (due to the decrease in the weight of the propellant owing to consumption). In some mission, the thrust required will be constant and some missions require the thrust to be increased with altitude (Moving from one orbit to other, boosting is required).

The most popular and time tested method of controlling thrust on a SPR is by varying the geometry of the solid propellant grain. We know that thrust obtained is directly proportional to the surface area exposed to combustion. Larger the surface area exposed to combustion, larger the exhaust gases passing through the combustion chamber and the nozzle, higher will be the thrust obtained and vice versa. This is referred as **instantaneous burning area A_b** . So by controlling the instantaneous burning area, the thrust can be controlled or altered and this is known as **internal ballistics**. The branch of applied science describing the burning rate, burning surface, and grain geometry is known as **internal ballistics**.

During the combustion, surface area of the propellant recedes in the direction normal (perpendicular) to the surface at that point and due to the consumption of the grain, a new surface area will be evolved. The newly evolved surface area will be slightly different from the previous surface area and the cross sectional area of the grain will also be different. The cross sectional area of the grain is an important factor which affects the performance of the rocket. If the newly evolved surface area has a larger surface area, the mass flow rate of the combustion gases increases which ultimately increases the thrust of the rocket.

Let us consider a two-dimensional cylindrical cavity with 5 slots as shown in figure. Upon ignition, combustion will be initiated along all the slots of the cavity and the propellant grain will recede in a direction normal to its surface. Initially the slots are sharp, but as time grows and at different time intervals, a new shape with a different surface area will evolve as seen from the figure. With the newly formed grain surface area, the thrust generated will proportionally be affected.



Figure: Successive burning surface contours, each a fixed small time apart

The mass flow rate of the exhaust gases is given by the expression, where A_b is the burning area of the propellant grain, r the burning rate, and ρ_b the solid propellant density prior to motor start. The total mass m of effective propellant burned can be determined by integrating the equation.

$$\dot{m} = A_b r \rho_b \quad (1)$$

$$m = \int \dot{m} dt = \rho_b \int A_b r dt \quad (2)$$

In the equation, A_b and r can be varied with time and pressure to extensively understand the burning characteristics of the propellant grain. The burn rate of any propellant is further dependent on the chamber pressure, given by the empirical relation

$$r = a p_1^n \quad (3)$$

where r , the burn rate, is usually in centimetres per second or inches per second, and the chamber pressure P_1 is in MPa or psia, a is an empirical constant influenced by ambient grain temperature. Also a is known as the **temperature coefficient** and it is not dimensionless. The **burning rate exponent n** , sometimes called the **combustion index**, is independent of the initial grain temperature and describes the influence of chamber pressure on the burning rate.

Thrust Profile of a Propellant Grain Geometry

Thrust profile, sometimes referred as thrust curve is defined as the graph of the thrust of an engine or motor with respect to time. Various thrust profiles can be obtained for a given rocket mission or application. The obtained thrust profile completely depends on the grain geometry of the solid propellant and the propellant

mass flow rate as given in eqn 1. Basically, by changing the geometry of the propellant, thus change the propellant mass flow rate, three different types of thrust profiles can be achieved. They are

1. Regressive Burning
2. Neutral Burning
3. Progressive burning

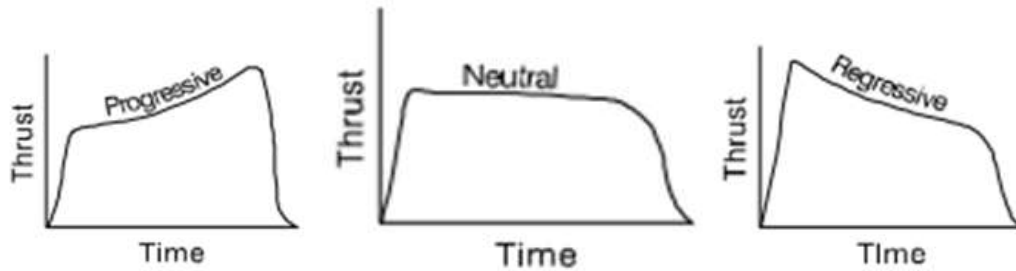


Figure: Progressive, neutral and regressive thrust profiles for solid propellants grains

Progressive Burning- In progressive burning, the surface area A_b of the propellant increases during the interval of combustion. Due to this, propellant mass flow rate increases and the overall thrust increases with time. A simple configuration that exhibits progressive burning is a case-bonded grain with a cylindrical core. As burning proceeds, the diameter of the core increases, causing the burning surface area to increase, resulting in increased gas production and thrust. A sample progressive burning thrust profile is provided in figure below.

Neutral Burning- In neutral burning, the surface area A_b of the propellant remains constant during the interval of combustion. Due to this, both propellant mass flow rate and the thrust produced remains constant with time. Many propellant grain geometries can be configured to approximate neutral burning. The thrust profile for a simple, end-burning propellant grain is illustrated below, where the slight increase in thrust (typically a temperature effect) is ignored.

Regressive Burning- In regressive burning, the surface area A_b of the propellant reduces during the interval of combustion. Consequently, the propellant mass flow rate too decreases causing the thrust production also to recede over time.

Various propellant grains with their thrust profile are describe and depicted in the figure.

- (1) **Solid Propellant with cylindrical channel-** In such configurations, the surface area increase with time and hence the thrust also increases with time. (Progressive Burning-Used for Terminal Staging)
- (2) **Solid Propellant with a channel and a central cylinder-** Such geometry produce a relatively constant thrust, which reduces to zero very quickly when the fuel is used up. (Neutral Burning- Mainly used for Lift Off)
- (3) **Solid Propellant with star shaped geometry-** Develops a relatively constant thrust which decreases slowly to zero as the last of the fuel is consumed. (Neutral Burning- Mainly used for Lift Off)
- (4) **Solid Propellant with cruciform geometry** - The 'cruciform' profile produces progressively less thrust. (Gradual Regressive Burning- Mainly used for Orbital Insertion)
- (5) **Solid Propellant with double anchor profile** - Produces a decreasing thrust which drops off quickly near the end of the burn. (Instantaneous Regressive Burning- Mainly used for Missiles)
- (6) **Solid Propellant with wagon wheel profile** - produces a strong initial thrust, followed by an almost constant lower thrust. (Regressive Burning- Mainly used for Missiles)

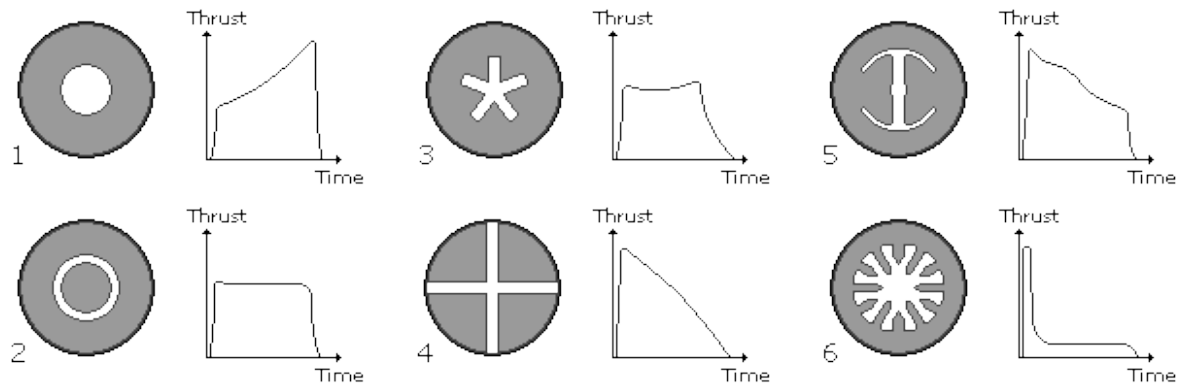


Fig : Thrust profiles for various grain geometries

Ignition and Combustion Process

Ignition and combustion are two very important processes in any thrust producing machines. These two processes are strongly intertwined to each other wherein ignition will always precede the process of combustion.

The first process in the combustion of propellant grain is ignition, and ignition always precedes combustion process and there is no combustion process without ignition. The process of ignition itself consists of a series of complex rapid events that immediately start at the receipt of an electric signal. The various processes that occur after ignition are

- **Heat generation** - The igniter generates the required heat for igniting the propellant grain and in most of the cases the ignition process will not take more than fraction of a second. Once ignition occurs, the next immediate process is increase of the pressure in the port cavity. The combustion chamber pressure rises from a null state to an equilibrium state quickly as seen in the figure. **Heat generation** - The igniter generates the required heat for igniting the propellant grain and in most of the cases the ignition process will not take more than fraction of a second. Once ignition occurs, the next immediate process is increase of the pressure in the port cavity. The combustion chamber pressure rises from a null state to an equilibrium state quickly as seen in the figure. For the combustion process to sustain, an equilibrium pressure with full gas flow has to be maintained which will be responsible in transferring energy from the igniter gas to the grain surfaces.
- **Transfer of the heat from the igniter to the motor grain surface** - The heat generated by the igniter will be quickly transferred to the grain surfaces.
- **Spreading the flame over the entire burning surface area** – Here, the flame will quickly develop and instantly spread over the entire surface area that is exposed to combustion and covers maximum surface area of the grain.
- **Filling the port cavity with gas** - With the burning of the propellant grain, the mass flow rate of the exhaust gases tremendously increases and completely fills the entire port cavity of the motor. With this, the pressure in the thrust chamber tremendously increases. For the combustion process to sustain, an equilibrium pressure with full gas flow has to be maintained which will be responsible in transferring energy from the igniter gas to the grain surfaces.

Conventionally, the ignition process is divided into three phases for analytical purposes:

Phase I-Ignition time lag: the period from the moment the igniter receives a signal until the first bit of grain surface burns.

Phase II-Flame-spreading interval: the time from first ignition of the grain surface until the complete grain burning area has been ignited.

Phase III-Chamber-filling interval: the time for completing the chamber filling process and for reaching equilibrium chamber pressure and flow.

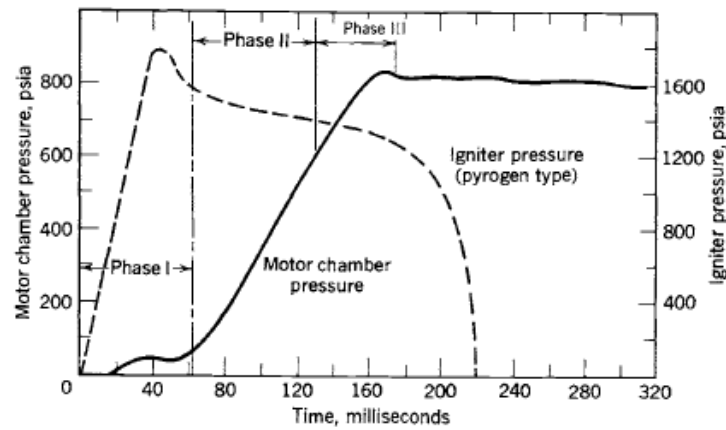


Figure: Ignition process showing different phases of ignition

Thrust Extinction or Thrust Termination

The thrust termination or extinction system is also an important system on board through which the rocket motor will be shut down. This is primarily important when the motor malfunctions and sometimes it's necessary to stop or extinguish the burning of the motor before all the propellant has been consumed. Hence, the thrust extinction system forms a failsafe mechanism and this forms an important control for a SPR. The problem with thrust termination system is that when activated it will not only shutdown the rocket but destroys the motor too and do not have any relighting or re-ignition possibility. The various circumstances which requires thrust termination are

1. When a flight vehicle has reached a desired flight velocity-It can be the predetermined velocity for a ballistic missile or a satellite which has reached the required orbit. The thrust has to be terminated, if not the vehicle will overshoot the given orbit or altitude.
2. Achieving thrust termination is very much important during flight testing of rockets or launch vehicles or missile. In case the test vehicle unexpectedly flies out of the safe boundaries, then it has to be immediately terminated.
3. To avoid collisions of stages during a stage separation manoeuvre (requiring a thrust reversal) for multistage flight vehicles.
4. During research and development testing, when one wants to examine a partially burned motor.

To achieve thrust termination, several methods are available. They are

a) Rapid depressurization-Simplest method of terminating the thrust is by reducing the combustion chamber pressure below deflagration limit. Deflagration limit is the minimum pressure at which combustion can still be barely self-sustained and maintained without adding energy. This type of depressurization occurs very slowly. Many solid propellants have a low-pressure combustion limit of 0.05 to 0.15 MPa. We know that the burn rate of a propellant is very much dependant on the chamber pressure given by the equation

$$\dot{m} = A_b r \rho_b$$

$$r = a p_1^n$$

Higher the pressure p_1 , faster will be the propellant burn rate r . Likewise, lower the chamber pressure p_1 , the burn rate r decreases accordingly. This means that to make the thrust zero, the pressure also needs to be reduced to zero. This can be done by rapidly reducing the chamber pressure either increasing the nozzle throat area or rapid opening of gas escape routes. Fast opening of the gas escape routes is commonly used on many motors. For this, usually blow out devices are placed at the forward end of the motor. Once activated, the devices explode, thus creating additional passages at the forward end through which the gases escape and neutralise the thrust. When the exhaust ports are opened, they not only allow the fast moving exhaust gases and reduce pressure but also help in decelerating the vehicle. Such a thrust reversal using ports located on the forward bulkhead of the case is achieved in the upper stages of Minuteman and Poseidon missiles. The location of the blow out devices is carefully selected to prevent any unbalanced forces being established on the vehicle. Generally, equal numbers of thrust termination blow out devices are located symmetrically opposite (2 or more) to each other. The forward flow of gas occurs only for a very brief period of time, during which

the thrust is actually reversed. The rapid depressurization causes a sudden stopping of the combustion at the propellant burning surface. With proper design the explosive cords do not cause a detonation or explosion of the remaining unburned propellant.

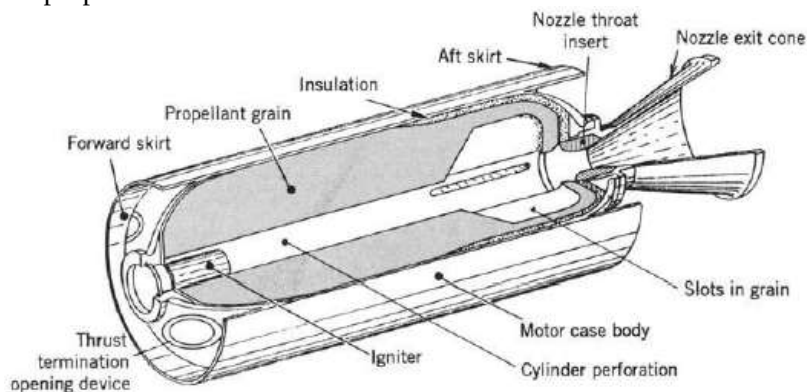


Figure: Solid propellant motor showcasing thrust terminating devices

b) Quenching with Water-Sometimes, studying the microstructure of a partially burnt propellant will be helpful especially while developing new propellant combination. For that purpose, the developed grain will be combusted and the thrust will be terminated halfway by injecting flame extinguishing liquid like water. This is one of the methods for thrust extinction specifically during research and development and not on an actual rocket.

Igniters for Solid Rocket Motors

The function of the igniter is to induce this combustion reaction in a controlled and predictable manner and at a stipulated rate.

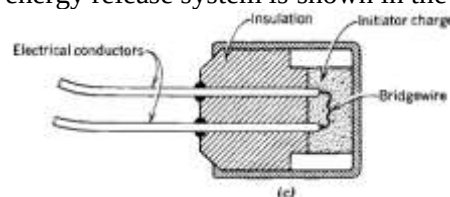
An ignition system for a solid rocket motor typically consists of 3 important systems,

- 1) An initiation system,
- 2) An energy release system, and
- 3) Associated hardware casing that encapsulates both ignition and the energy release system with the necessary mountings.

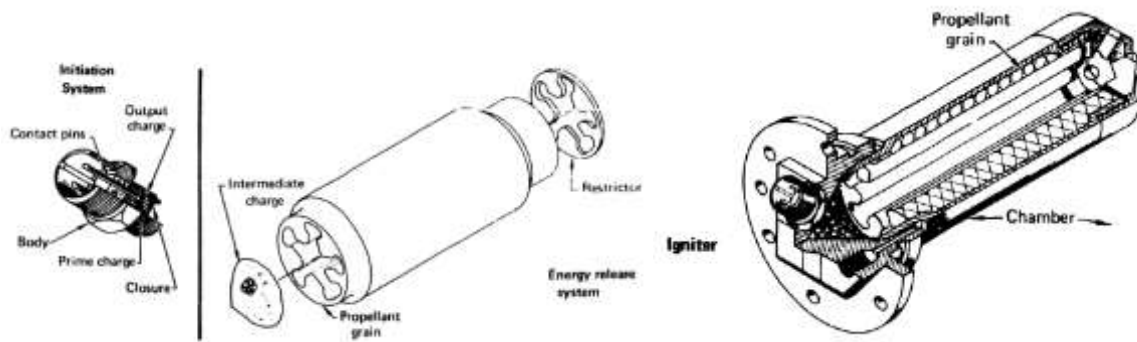
Initiation System- As the name implies, the initiation system provides a stimulus that actuates the motor ignition process. To initiate the ignition process, a mechanical, electrical, or chemical input can be used to initiate an energy release process.

Energy release system- The stimulus from the initiation system will be used to initiate the energy release system. The energy release system supplies the required energy, normally heat, required to ignite the propellant in the rocket motor.

Once the initiation system provides adequate ignition energy or the required stimulus, heat will be released which will ignite the energy release system. The heat energy released from the energy release system will ignite the motor propellant grain surface and an exothermic chemical reaction is initiated. As the exothermic chemical reactions become dominant, the temperature of the entire exposed propellant grain starts due to the heat transfer phenomenon through convection and radiation. As the propellant exothermic reactions rapidly become a dominant heat source, complete ignition of the grain is said to be accomplished. The ignition system encompassing the initiation and the energy release system is shown in the figure.



Igniter Squib



Types of Initiation Systems

1. Electroexplosive Devices

This is one of the most frequently used techniques to initiate energy. An Electroexplosive Device typically consists of 2 electrically insulated terminals that ends into an explosive chemical compound packed in a small case. The terminal with the chemical charge is commonly referred to an initiator” or “squib. The terminals are connected to an electrical circuit or a voltage source and when the required current is applied, a high intensity spark is discharged at the terminals. The high intensity and high temperature spark will immediately ignite the chemical charge. Electroexplosive devices can conveniently be placed in one of two categories on the basis of the energy source required for initiation: (1) low-voltage devices are fired by direct application of current from a voltage source usually of **28** volts or less; (2) high-voltage devices require a large potential (e.g., **500** to **2500** volts) to initiate the primary composition. Currently available well-designed EED’s provide highly reproducible ignition times of less than 5 msec at recommended firing levels.

2. Bulkhead Devices

To eliminate some of the safety problems associated with electroexplosive devices, a method of initiating solid rocket igniters has been developed in which a detonation shock wave is transmitted from a donor charge through a solid metal interface to an acceptor charge that initiates the deflagration of a heat-producing composition.

3. Mechanical Devices

Mechanical devices are not commonly used for initiating ignition energy. A mechanical ignition device generally consists of a metal cup into which an impact-sensitive chemical charge, or primer mix, is loaded. The charge is retained in place with a thin disc, and an anvil is inserted so that the impact of a firing pin against the anvil will ignite the entrapped primer mix which will create the ignition of the energy release system.

4. Laser

Sufficient testing has been done to demonstrate that light energy, through the laser system, can be used to initiate solid rocket igniters.

5. Hypergolic

Certain oxidizers, e.g., Chlorine Trifluoride, react hypergolically on contact with solid rocket propellants. These oxidizers have been used to initiate pyrogens and sustain combustion of pyrogens containing fuel only, and have found some use as primary motor igniters. The oxidizers usually are injected under pressure into the combustion chamber in liquid or gaseous form.

Energy Release System

The energy release system is that portion of the solid rocket igniter that provides the heat efflux necessary to ignite the motor propellant and raise it to a self-sustaining combustion level. A wide variety of igniter “types” has been developed. However, currently, only 3 types of igniters are being used.

1. Pyrotechnic Igniters
2. Pyrogen Igniters
3. Hypergolics

Pyrotechnic Igniters or Pelleted Pyrotechnics

Pyrotechnic igniters are commonly used in small rocket motors. They are less powerful but still effective in initiating the combustion process. **Pyrotechnic** igniters usually use solid explosives or energetic propellants as heat-producing material. The solid explosives are in the form of small pellets or small grains which are packed and stored in a bag or can or basket, perforated tube, combustible case, jellyroll, etc. Among all, only the pelleted type igniters are most commonly used. The pellets are stored in a metallic basket and act as the main charge. The chemical composition of the pellets is 24% boron-71% potassium perchlorate-5% binder.

In the pyrotechnic igniter, as a standard construction and operation, it consists of an initiation source which is nothing but the igniter squib or the primer charge. The igniter squib is connected to an electrical connector connected to an electric source and is surrounded by a small pack of sensitive chemical charge or the primer charge. Next, the igniter squib is in contact with a booster charge which is in direct contact with the perforated metallic basket consisting of tightly packed pellets. The igniter squib is similar to an electric match. During operation, first on receipt of an electrical signal the initiator releases the energy by igniting the squib. Later, the booster charge is ignited by heat released from the squib; and finally, the highly reactive pyrotechnic composition in the metallic basket is ignited and combustion is initiated. These pellets give a large burning surface and a short burning time. Once ignited, the pyrotechnic charge produces a high-temperature flame causing the pressure to build up in the combustion chamber. Once done, the combustion process sustains and continues.

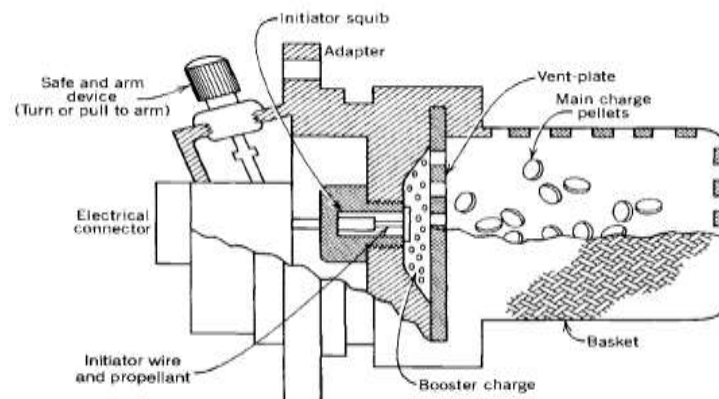


FIGURE 14-14. Typical pyrotechnic igniter with three different propellant charges that ignite in sequence.

Pyrogen Igniters

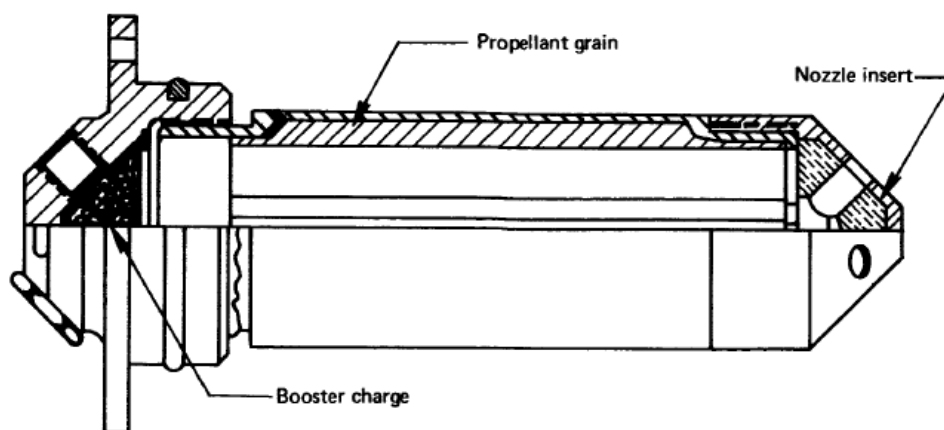
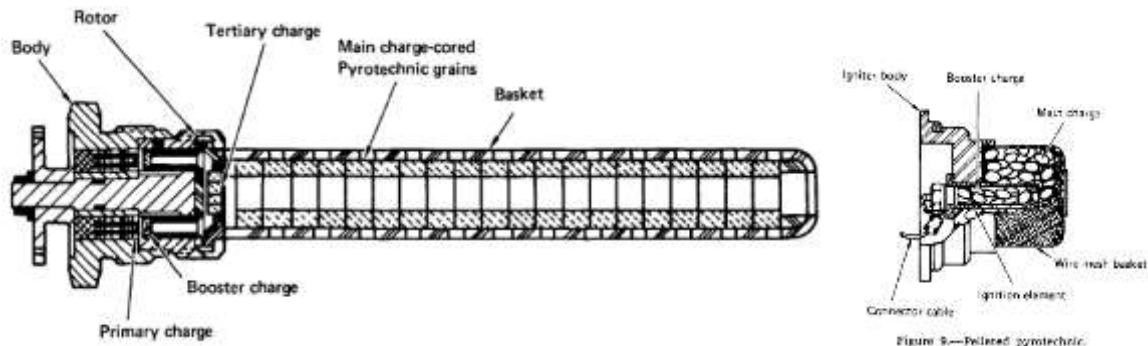


Figure 7.—Pyrogen igniter.

Pyrogen Igniters are slightly complex compared to the Pyrotechnic Igniters. These igniters are commonly used in large rocket motors and ballistic missiles. Since the volume of the propellant is enormous in large scale rockets and missiles, a higher heat efflux is required to ignite the motor grain and hence, the pyrogen igniters are commonly used. These igniters liberate more heat energy that is generally required to ensure reliable ignition. The pyrogen igniter is nothing but a small rocket motor itself and this small motor is used to ignite a larger rocket motor. In most cases, the small igniter motor and the large propellant grain are made of the same

compositions. A common igniter formulation uses 20 to 35% boron and 65 to 80% potassium nitrate with 1 to 5% binder. Binders typically include epoxy resins, graphite, nitrocellulose, vegetable oil, polyisobutylene, and other binders. Another formulation uses magnesium with a fluorocarbon (Teflon). In most cases, a pyrotechnic igniter will itself be used as an igniter in the pyrogen. Since the igniter propellant mass is small (often less than 1% of the motor propellant) and burns mostly at low chamber pressure (low Is), it contributes very little to the motor overall total impulse.



In this type of ignition system, the igniter consists of an electrical squib packed with a booster charge as shown in figure. The boost charge is usually a readily ignited pelleted pyrotechnic charge. Upon electrical ignition, the booster charges get ignited and the heat energy from the booster charges readily ignites the pelleted pyrotechnic and propagates the ignition train from the initiator to the pyrogen propellant grain. Reaction products from the pyrogen grain are expelled through the pyrogen nozzle and impinge on the surface of the motor propellant, thus initiating a self-sustained exothermic chemical combustion reaction process. Heat transfer from the pyrogen to the motor grain is largely convective, with the hot gases contacting the grain surface as contrasted to a highly radiative energy emitted by pyrotechnic igniters. The combustion process continues until the deflagration limit is reached, beyond which the combustion ceases and the thrust production stops.

Hypergolic

The hypergolic ignition technique consists of spraying a hyperbolic liquid oxidiser that reacts exothermically when the propellant is contacted. Although not currently used in operational systems, hypergolics have sufficient potential for multiple restart ignitions to warrant a brief review of current design practices. Chlorine trifluoride in liquid form is the hypergolic reagent used for most propellants. This Chlorine Trifluoride liquid acts as an oxidiser and the solid propellant acts as a fuel. The hypergolic oxidiser is injected into the motor port at a very high pressure using efficient nozzle to produce maximum dispersion and drop size reduction. By using the nozzle, initially the hypergolic liquid is impinged upon the forward half of the motor port. Upon contact, combustion starts and the thrust production is initiated. A mixture of perchlorylfluoride and chlorine trifluoride has been used successfully.

Liquid Propellant Rockets

This section is completely devoted to liquid propellant rockets and gives an overview of the all the processes that occurs in a LPR. It presents the different principal sub systems of LPR like the basic working, propellant types, feed systems, Igniters, cooling systems etc. Before diving into these concepts, as a customary requirement let us first understand in detail the working of a LPR.

A rocket which uses liquid propellants, i.e., liquid fuel and liquid oxidizer for propulsion is called as Liquid propellant rocket. A liquid propellant rocket propulsion system is commonly called a rocket engine. It has all the hardware components and propellants necessary for the task of producing the required thrust. The most important part of any liquid propellant rocket is the thrust chamber because it provides housing for major components like the combustion chamber itself, propellant tanks, feed mechanisms, fuel injectors and other control devices to initiate and regulate the propellant flow. Based on the mission requirement, a typical rocket engine can have 1 or more thrust chambers.

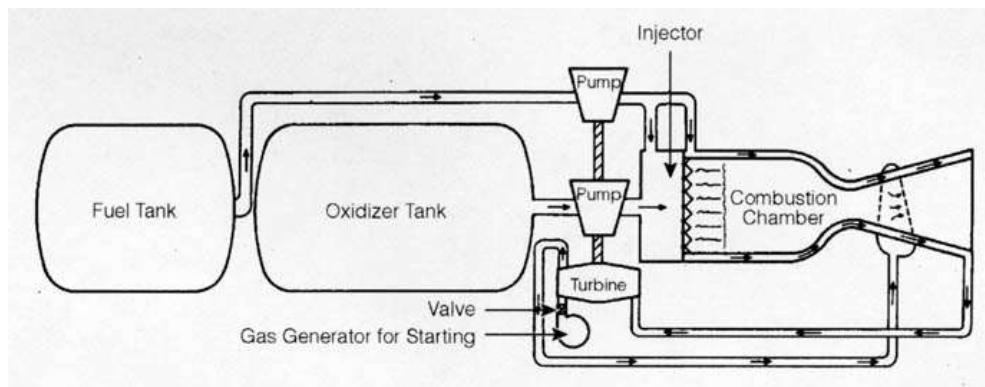


Figure: Liquid Propellant Rocket Schematic

In liquid propellant rockets, the liquid fuel and liquid oxidizer will be stored in separate tanks at a very high pressure (30bar). In the most widely used bipropellant system two propellants (fuel and oxidizer) will be used and hence they require two individual storage tanks. During the working of any typical liquid propulsion rockets, the liquid propellants are introduced into the thrust chamber at very high pressure through propellant feed systems. Usually the propellants are introduced using either the pump fed systems or pressure fed system depending on the configuration of the engine. Pump fed systems are mainly used to force the propellants from the tanks to the thrust chamber through a well-designed plumbing or piping system. For better mixing and efficient combustion, the liquid propellants are metered, injected, atomized and mixed thoroughly by passing through the fuel injectors. After thorough mixing is achieved, the fuel-oxidizer charge is ignited electrically using igniters which results in combustion of propellants. The energy from a high-pressure combustion reaction of propellant chemicals, a fuel and an oxidizing chemical, permits the heating of reaction product gases to very high temperatures (2500 to 4100°C). These gases subsequently are expanded in a supersonic nozzle and accelerated to high velocities (1800 to 4300 m/sec) imparting a huge thrust. In some applications an engine may also include a thrust vector control system to provide additional directional control to the rocket.

Since these gas temperatures are about twice the melting point of steel, it is necessary to cool or insulate all the surfaces that are exposed to the hot gases. In a cooled thrust chamber, one of the propellants (usually the fuel) is circulated through cooling jackets or a special cooling passage to absorb the heat that is transferred from the hot reaction gases to the thrust chamber walls (see Figs 8-2 and 8-3). A radiation-cooled thrust chamber uses a special high-temperature material, such as niobium metal, which can radiate away its excess heat. There are uncooled or heat-absorbing thrust chambers, such as those using ablative materials.

Advantages

1. Liquid Propellant engines can be reused after recovery. This offers considerable economic benefits. Space Shuttle main engine
2. They offer more flexibility and greater control over the thrust. The operation is more stable.
3. Higher values of specific impulse.
4. Wreckage on account of some defect or malfunctioning is not sudden allowing time for corrective measure and safety action. This offers greater safety to the crew.
5. It is much easier to stop the operation in case of an impending catastrophe.
6. Regenerative heating of the propellant before combustion along with the cooling of the nozzle and the thrust chamber is possible. This offers thermodynamic and structural advantages.
7. More economical for long range space and military operations.

Disadvantages

1. More complicated compared to solid rockets. There is greater probability of problems arising on account of malfunctioning of turbo-pump systems, valves, leakage, vibrations etc.
2. Many liquid propellants are cryogenic which require special arrangements for heat insulation, solar radiation shields and handling at low temperatures.
3. There are additional handling and safety problems if the propellants are poisonous and hazardous.

4. They require propellant feed pumping systems or tank pressurization increasing the total weight of the rocket. A liquid propellant rocket will burn nearly 4000 liters of propellant per second and just imagine how sophisticated the components should be.
5. On account of lower density of the liquid propellants compared to solid propellants, large storage space is required thus increasing the size and weight of the engine.
6. It takes much longer time to design, manufacture and test liquid propellant rockets. They cannot be mass produced in a short time for military applications.

Liquid propellant rocket engine systems can be classified in several other ways. They can be reusable (like the Space Shuttle main engine or a booster rocket engine for quick ascent or maneuvers of fighter aircraft) or suitable for a single flight only (as the engines in the Atlas or Titan launch vehicles) and they can be restartable, like a reaction control engine, or single firing, as in a space launch vehicle. They

1. Mostly used for primary propulsion engines
2. Used in lower stages of most of the rockets
3. Upper stages of missiles
4. In deep space rockets

Propellant Properties

Propellants are the working substances of rocket engines that are usually fluid in nature. When ignited, the propellants usually undergo chemical and thermodynamic changes that aids in producing thrust. The important properties, performance, and characteristics of common liquid propellants are stated below. These characteristics affect the engine design, test facilities, propellant storage and handling.

Properties of Liquid Propellants

1. Energy released during combustion per unit mass of the propellant combination should be high.
2. High density propellants are preferred; they require smaller tanks and structures offering lower values of the mass ratio and aerodynamic drag.
3. The propellants should have lower freezing point; this makes the handling of propellants, particularly cryogenic propellants easier in various rocket components. Low freezing point rules out any possibility of freezing of the propellants at high altitudes and in cold weather.
4. They must be non-corrosive so that their handling is easier and less demanding on materials used in various parts of the rocket.
5. They should be chemically stable. Their properties should not deteriorate with time. They should not absorb moisture. Small amounts of impurities should not change their chemical properties.
6. Propellants should have low values of vapor pressure and viscosity. Lower vapor pressure avoids Cavitation problems in line feed pumping system. Lower viscosity decreases the power required for pumping.
7. Propellants must have higher specific heat and thermal conductivity for better performance.
8. They should not be poisonous and hazardous. Some propellants are poisonous; they are dangerous for personnel if inhaled and touched. Others ignite easily when they come in contact with air. Such propellants may explode if not carefully stored and handled. Handling, transportation and storage should be simple, convenient and safe.
9. Propellants chosen should be cheap and abundantly available. The raw materials required for their large scale manufacture should be easily available. The manufacturing process required for producing propellants should be simple and inexpensive.

Types of Liquid Propellants Used for Rocket Engines

Many different types of storable and cryogenic liquid oxidizer propellants have been synthesized, proposed and implemented. The types of liquid propellants are

Liquid Monopropellants

A monopropellant contains an oxidizing agent and combustible matter in a single substance. It may be a mixture of several compounds or it may be a homogeneous material, such as hydrogen peroxide or hydrazine. Monopropellants are stable at ordinary atmospheric conditions but decompose and yield hot combustion gases when heated or catalysed. The propellant-feed and control-system simplicity associated with a monopropellant makes this type of propellant attractive for certain applications. Hydrazine is being used extensively as a monopropellant in small attitude and trajectory control rockets for the control of satellites and other spacecraft and also as a hot gas generator. Concentrated hydrogen peroxide was used for monopropellant gas generation in the USA, Russia, and Germany

Hydrazine

Hydrazine is not only an excellent storable fuel, but also an excellent monopropellant when decomposed by a suitable solid or liquid catalyst; this catalyst often needs to be preheated for fast startup. Iridium is an effective catalyst at room temperature. At elevated temperature (about 450 K) many materials decompose hydrazine, including iron, nickel, and cobalt. As a monopropellant, it is used in gas generators or in space engine attitude control rockets.

Hydroxyl Ammonium Nitrate (NH₂OH*NO₃)

This is a relatively new, synthetic, propellant material rich in oxygen, but with combined hydrogen and nitrogen (fuel ingredients), it is abbreviated as HAN. It is an opaque hygroscopic solid when pure, and a clear colorless odourless liquid in aqueous solutions. The solid HAN (specific gravity of 1.84) is a potential solid propellant ingredient and the liquid HAN solution is a potential monopropellant (a 13 molar solution has a specific gravity of 1.523).

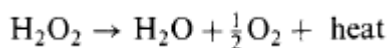
Bipropellants

Oxidizer

Liquid Oxygen (O₂) Liquid oxygen, often abbreviated as LOX, boils at 90 K at atmospheric pressure; at these conditions it has a specific gravity of 1.14 and a heat of vaporization of 213 kJ/kg. It is widely used as an oxidizer and burns with a bright white-yellow flame with most hydrocarbon fuels. It has been used in combination with alcohols, jet fuels (kerosene-type), gasoline, and hydrogen. As shown in Table 5-5, the attainable performance is relatively high, and liquid oxygen is therefore a desirable and commonly used propellant in large rocket engines. The following missiles and space launch vehicles use oxygen: (1) with jet fuel-- Atlas, Thor, Jupiter, Titan I, Saturn booster; (2) with hydrogen--Space Shuttle and Centaur upper stage; (3) with alcohol--V-2 and Redstone. Figures 1-4 and 6-1 show units that use oxygen.

Hydrogen Peroxide (H₂O₂)

In rocket application, hydrogen peroxide has been used in a highly concentrated form of 70 to 99%; the remainder is mostly water. Commercial peroxide is approximately 30% concentrated. Concentrated hydrogen peroxide was used in gas generator and rocket applications between 1938 and 1965. In the combustion chamber, the propellant decomposes according to the following chemical reaction, forming superheated steam and gaseous oxygen:



This decomposition is brought about by the action of catalysts such as various liquid permanganates, solid manganese dioxide, platinum, and iron oxide.

Nitric Acid (HNO₃)

There are several types of nitric acid mixtures that have been used as oxidizers. The most common type, red fuming nitric acid (RFNA), consists of concentrated nitric acid (HNO₃) that contains between 5 and 20% dissolved nitrogen dioxide. The evaporating red-brown fumes are exceedingly annoying and poisonous.

Compared to concentrated nitric acid (also called white fuming nitric acid), RFNA is more energetic, more stable in storage, and less corrosive to many tank materials. Nitric acid is highly corrosive. Only certain types of stainless steel, gold, and a few other materials are satisfactory as storage containers or pipeline.

Nitrogen Tetroxide (N₂O₄)

Nitrogen tetroxide is a storable propellant oxidizer and is used in the Titan missile together with a fuel mixture consisting of hydrazine and unsymmetrical dimethylhydrazine. It is also used with monomethylhydrazine fuel in the Space Shuttle orbital maneuver system and reaction control system and in many spacecraft propulsion systems.

Liquid Fuels

Hydrocarbon Fuels: Petroleum derivatives encompass a large variety of different hydrocarbon chemicals, most of which can be used as a rocket fuel. Most common are those types that are in use with other applications and engines, such as gasoline, kerosene, diesel oil, and turbojet fuel. They are relatively easy to handle, and there is an ample supply of these fuels available at low cost. A specifically refined petroleum product particularly suitable as a rocket propellant has been designated RP-1.

Methane (CH₄): Methane (CH₄) is a cryogenic hydrocarbon fuel. It is denser than liquid hydrogen and relatively low in cost. Compared to petroleum refined hydrocarbons it has highly reproducible properties. With liquid oxygen it is a candidate propellant combination for launch vehicle booster rocket engines and also reaction engines control when oxygen is available from the main engines.

Liquid Hydrogen (H₂)

Liquid hydrogen, when burned with liquid fluorine or liquid oxygen, gives a high performance, as shown in Table 5-5. It also is an excellent regenerative coolant. With oxygen it burns with a colorless flame; Of all known fuels, liquid hydrogen is the lightest and the coldest, having a specific gravity of 0.07 and a boiling point of about 20 K. The very low fuel density requires bulky fuel tanks, which necessitate very large vehicle volumes. The extremely low temperature makes the problem of choosing suitable tank and piping materials difficult, because many metals become brittle at low temperatures.

Hydrazine (N₂H₄)- It is used as a bipropellant fuel as well as a monopropellant. Hydrazine and its related liquid organic compounds, monomethylhydrazine (MMH) and unsymmetrical dimethylhydrazine (UDMH), all have similar physical and thermochemical properties. Hydrazine is a toxic, colorless liquid with a high freezing point (274.3 K).

Unsymmetrical Dimethylhydrazine [(CH₃)₂NNH₂] A derivative of hydrazine, namely, unsymmetrical dimethylhydrazine (UDMH), is often used instead of or in mixtures with hydrazine because it forms a more stable liquid, particularly at higher temperatures. Furthermore, it has a lower freezing point (215.9 K) and a higher boiling point (336.5 K) than hydrazine. When UDMH is burned with an oxidizer it gives only slightly lower values of *I_s* than pure hydrazine. This fuel is used in the Titan missile and launch vehicle and spacecraft engines in 50% mixtures and has been used in the lunar landing and take-off engines. UDMH is used in Russian and Chinese rocket engines.

Monomethylhydrazine (CH₃NHNH₂)

Monomethylhydrazine (MMH) has been used extensively as a fuel in spacecraft rocket engines, particularly in small attitude control engines, usually with N₂O₄ as the oxidizer. It has a better shock resistance to blast waves, better heat transfer properties, and a better liquid temperature range than pure hydrazine.

Storable propellants (e.g., nitric acid or gasoline) are liquid at ambient temperature and can be stored for long periods in sealed tanks. Space storable propellants are liquid in the environment of space; this storability depends on the specific tank design, thermal conditions, and tank pressure. An example is ammonia.

Cryogenic Propellants

A cryogenic propellant is liquified gas at low temperature, such as liquid oxygen (-183°C) or liquid hydrogen (-253°C). Provisions for venting the storage tank and minimizing vaporization losses are necessary with this type.

Gelled propellant

A gelled propellant is a thixotropic liquid with a gelling additive. It behaves like a jelly or thick paint. It will not spill or leak readily, can flow under pressure, will burn, and is safer in some respects.

Cold Gas Propellant

A cold gas propellant (e.g., nitrogen) is stored at very high pressure, gives a low performance, allows a simple system and is usually very reliable. It has been used for roll control and attitude control.

Propellant Feed Systems

The propellant feed system has two principal functions: to raise the pressure of the propellants and to feed them to one or more thrust chambers. All feed systems have piping, a series of valves, provisions for filling and removing (draining and flushing) the liquid propellants, and control devices to initiate, stop, and regulate their flow and operation.

Pressure Feed System

One of the simplest and most common means of pressurizing the propellants is to force them out of their respective tanks by displacing them with high-pressure gas. This gas is fed into the propellant tanks at a controlled pressure, thereby giving a controlled propellant discharge. Because of their relative simplicity, the rocket engines with pressurized feed systems can be very reliable.

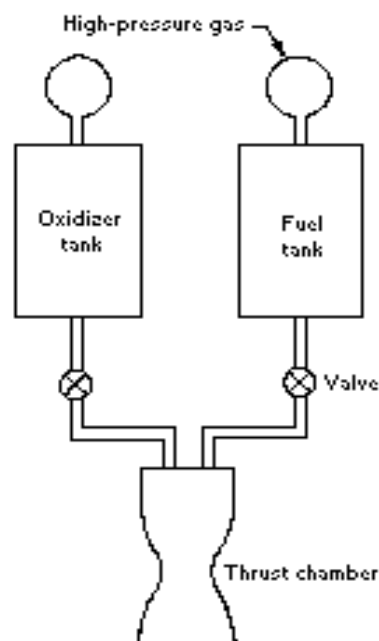


Figure: Pressure feed system

A simple pressurized feed system is shown schematically in the figure. It consists of a high-pressure gas tank, a gas starting valve, a pressure regulator, propellant tanks, propellant valves, and feed lines. Additional components, such as filling and draining provisions, check valves, filters, flexible elastic bladders for separating the liquid from the pressurizing gas, and pressure sensors or gauges, are also often incorporated. After all tanks are filled, the high-pressure gas valve is remotely actuated and admits gas through the pressure regulator at a constant pressure to the propellant tanks. The propellants are fed to the thrust chamber by opening valves. When the propellants are completely consumed, the pressurizing gas can also scavenge and clean lines and valves of much of the liquid propellant residue. With monopropellants the gas pressure feed

system becomes simpler, since there is only one propellant and not two, reducing the number of pipes, valves, and tanks.

Pressure feed systems can be found on the combined Space Shuttle Orbital Maneuver System (OMS) and the Reaction Control System (RCS). The OMS provides thrust for orbit insertion, orbit circularization, orbit transfer, rendezvous, deorbit, and abort. The RCS provides thrust for attitude control (in pitch, yaw, and roll) and for small vehicle velocity corrections or changes in almost any direction (translation maneuvers), such as are needed for rendezvous and docking; it can operate simultaneously with or separate from the OMS.

TABLE 6-3. Characteristics of the Orbital Maneuver System (OMS) and the Reaction Control System (RCS) of the Space Shuttle in One of the Aft Side Podes

Item	OMS	Primary RCS	Vernier RCS
Thrust (per nozzle) (lbf)	6000	870	25
Number of thrusters per pod	1	12	2
Thrust chamber cooling	Regenerative and radiation	Radiation cooling	
Chamber pressure, nominal (psi)	125	152	110
Specific impulse (vacuum nominal) (sec)	313	280 ^a	265 ^a
Nozzle area ratio	55	22-30 ^a	20-50 ^a
Mixture ratio (oxide/fuel mass flow)	1.65	1.6	1.6
Burn time, minimum (sec)	2	0.08	0.08
Burn time, maximum (sec)	160	150	125
Burn time, cumulative (sec)	54,000	12,800	125,000
Number of starts, cumulative (sec)	1000	20,000	330,000
Oxidizer (N ₂ O ₄) weight in tank (lb)	14,866	1464	
Fuel (MMH) weight in tank (lb)	9010	923	
Number of oxidizer/fuel tanks	1/1	1/1	
Propellant tank volume, each tank (ft ³)	90	17.9	
Ullage volume, nominal (full tank) (ft ³)	7.8	1.2-1.5	
Tank pressure, nominal (psi)	250	280	
Helium storage tank pressure (psi)	4700	3600	
Number of helium tanks	1	2	
Volume of helium tanks (ft ³)	17	1.76	

^aDepends on specific vehicle location and scarfing of nozzle.

Sources: NASA, Aerojet Propulsion Company and Kaiser Marquardt Company.

In general, a pressure feed system gives a vehicle performance superior to a turbopump system when the total impulse or the mass of propellant is relatively low, the chamber pressure is low, the engine thrust-to-weight ratio is low (usually less than 0.6), and when there are repeated short-duration thrust pulses; the heavy-walled tanks for the propellant and the pressurizing gas usually constitute the major inert mass of the engine system.

Pump or Turbopump Feed System

In the turbo pump feed system propellants from the tanks are pumped into the combustor by gas turbine driven centrifugal pumps. The turbine or turbines work on high pressure and temperature gas generated separately or tapped out from the main combustor.

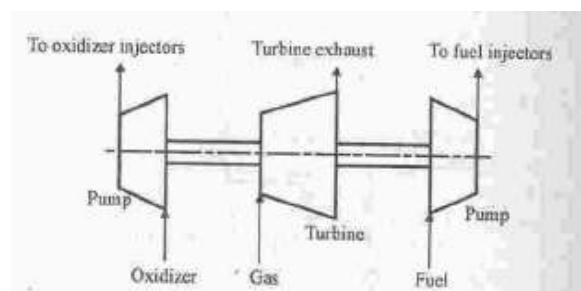


Fig : Turbo-pump system for liquid propellant rocket engines

The figure shows a general arrangement of a turbopump system. Here both the fuel and oxidizer pumps are driven by a single turbine. In order to achieve flexibility in choosing the design and operating parameters the fuel and oxidizer pumps can be driven separately by their turbines.

Figure depicts a turbopump feed system employing a single turbine driving the fuel and oxidizer pumps through a reduction gear. The turbine operates on a separate gas stream generated from the propellants in an independent gas generator as shown. A pressurizing gas can be used to increase the pressure of the propellants at the pump suction to avoid Cavitation and the resulting instability in pump operation.

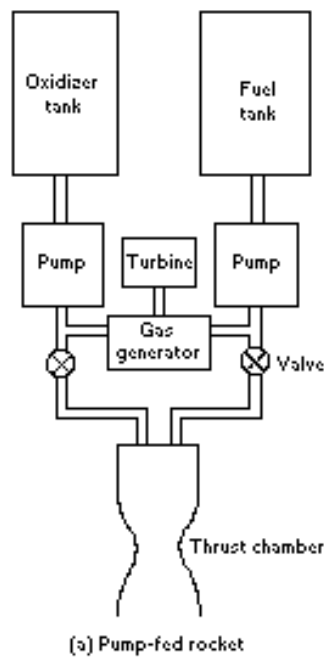


Fig : Turbopump feed system

Generally turbine speeds are high; therefore, propellant pumps can be driven at optimum speeds through reduction gears with an additional weight penalty. The working gas for the turbine can also be generated at the optimum temperature and pressure. The gas generator also has its own injection and ignition systems. The flow of propellants to the gas generator occurs (in the system shown in Figure) due to the action of the pressurizing gas. If gas pressurization is not employed the propellants can be bled from the delivery lines of the pumps. The Propellants flow required for driving the turbine is of the order of 1.5 to 5 per cent of the main flow. The turbine exhaust is also expanded through an exhaust nozzle to provide an additional thrust as shown in the figure. In a Turbopump feed systems the propellant tank pressures are much lower (by a factor of 10 to 40) and thus the tank masses are much lower (again by a factor of 10 to 40). Turbopump systems usually give a superior vehicle performance when the total impulse is large (higher Au) and the chamber pressure is higher.

Parameter	Thrust chamber	Engine
Specific impulse at sea level (max.), sec	368	362
Specific impulse in vacuum (max.), sec	421	415
Thrust, at sea level, lbf	640,700	650,000
Thrust in vacuum lbf	732,400	745,000
Mixture ratio	6.74	6.0

FIGURE 6-10. Simplified view of the RS-68 rocket engine with a gas generator cycle. For engine data see Table 10-3. (Courtesy of The Boeing Company, Rocketdyne Propulsion and Power.)

Injectors

An injector consists of an injector plate or a manifold on which finely drilled injection holes as shown in figure. These injection holes or orifices act as feed passages through which the high pressure fluid is converted to high velocity jets streams. A large number of orifices help in achieving a good distribution of flow over the cross section of the combustion chamber. This has been accomplished with different types of injector designs and elements.

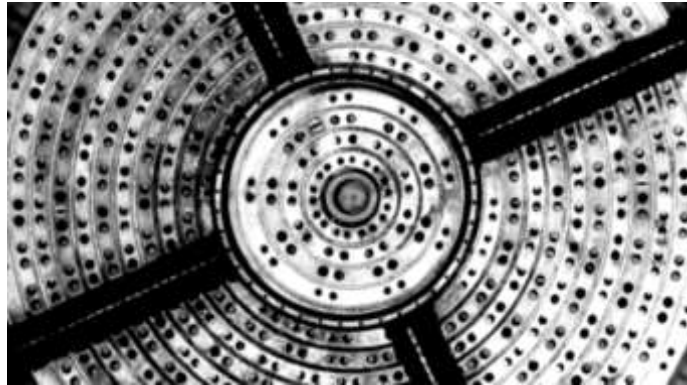


Fig: Manifold plate with numerous drilled holes

Non-impinging or Showerhead injector

The oldest type of injector is the showerhead. A showerhead will have a high pressure liquid propellant forced through an injector plate or a manifold with numerous finely drilled parallel holes or orifices. The high pressure fluid from storage tanks when passing through the narrow holes will experience a drop in pressure with an increase in velocity, resulting in multiple jets emerging normal to the face of the injector. These jets of fluid issuing out of the orifices will in turn break up into fine droplets at some downstream location and mix up quickly. It relies on turbulence and diffusion to achieve mixing.



Fig: Actual images of shower head injectors

In case of the showerhead type of injector, the injector produces narrow fluid jets, but the jets flow out in a straight line without diverging and rely on turbulence in the combustion chamber to enhance mixing process. This generally results in poorer mixing between the fuel and oxidiser. This can be compensated by increasing the combustion chamber length so that the jets coming out of the manifold will have sufficient time to mix purely through turbulence and diffusion. But this unnecessarily increases weight penalties. Hence they are restricted to hobby and amateur rockets.

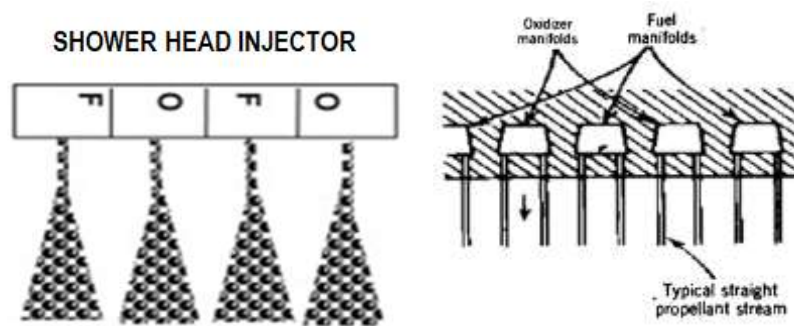


Fig: Schematic representation of shower head injectors

Impingement Injectors

Like-on-like or self-impinging (fuel-on-fuel and oxidizer-on-oxidizer)

As an improvement to the showerhead design, impingement injectors were designed and introduced. In impingement type of injectors, a pair of holes (doublets) is drilled throughout the injector plates at an angle opposite to each other through the manifold so that the propellant jets ejecting out at an angle will collide with each other. Later the colliding jets will scatter the fluid over a wider angle, thus atomizing and also enhancing the mixing between the two jets, and usually mixing happens downstream. Such injectors provide a significant improvement in atomization and mixing. For fuel and oxidiser, different diameter holes are drilled to satisfy the stoichiometric requirements (Bigger hole for Oxygen as it needs to be supplied in large quantities compared to fuel). Impingement injector is commonly used in ATLAS, TITAN, THOR and SATURN-V rockets as it was easier to design. This particular design had 2800 holes (Doublets/Pair) for both fuel and oxidiser.

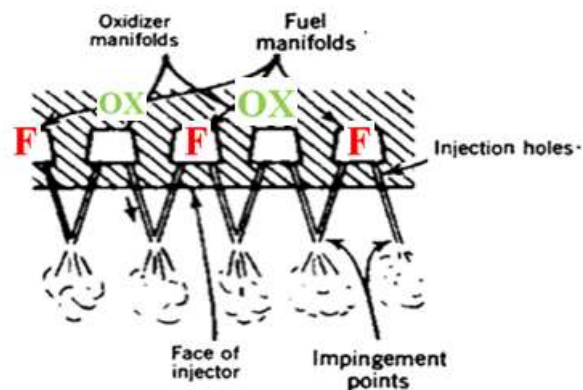


Fig: Illustration of the working of Like-on-like or self-impinging injector

Unlike Doublet (oxygen-hydrocarbon and storable propellants)

For unlike doublet patterns the propellants are injected through a number of separate small holes in such a manner that the fuel and oxidizer streams impinge upon each other. The two liquid streams then form a fan which breaks up into droplets. The major problem with impingement injectors is the holes need to be drilled with extreme accuracy especially at desired angle. If there is a mismatch in the angle between a pair of jets, then it will affect both atomization and mixing as they may not impinge on to each other properly. In such cases, if there is an error in jet angle, then the whole of the injector needs to be discarded.

Based on the impingement principle, a host of modifications to the geometry resulted in different types of impingement injectors. Basically for rocket application, impingement injectors can be either like impinging or unlike impinging injector. In the like impinging injector, the fluid jets will be of the same type of propellant (Oxidiser or fuel), whereas in unlike impinging injector, the jets can be of fuel and propellant combination.

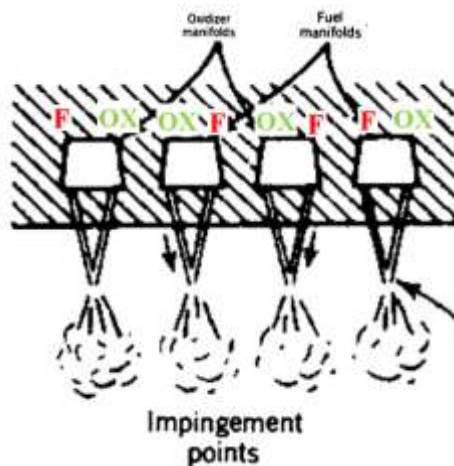


Fig: Self-impinging and Doublet impinging stream pattern

Triplet

For the same double impinging injector a third orifice was added in the middle of every doublet. Later, the like impinging injector was redesigned as an unlike impinging injector where in case of a doublet, each orifice supplied fuel and oxidiser. In such designs, the fuel oxidiser combination can immediately burn as soon as they are mixed. Since, the mass flow rate of oxidiser is always more than the fuel mass flow rate, the effective jet after mixing would normally tilt at an angle. To contain this problem, a triplet was added to the doublet configuration wherein the middle orifice would supply oxidiser and the remaining doublet pair would distribute fuel. This injector produces a symmetrical pattern with good mixing.

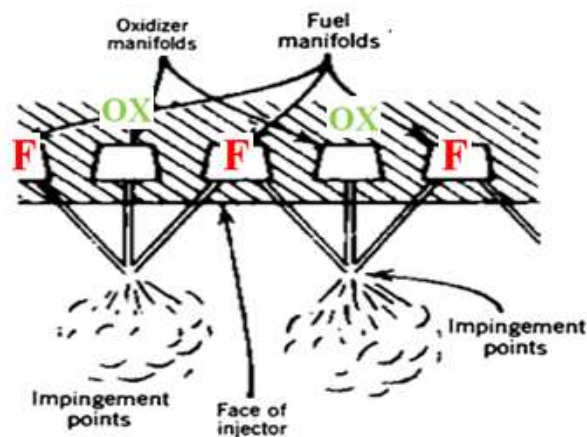


Figure: Triplet self-impinging injector

Spray Type Injectors

Latest type are the spray type injectors which produce conical spray which then spreads out break down as droplets. Best example for this is the swirl type injector in which two streams of fluid flows tangentially at an angle in the orifice. The holes are drilled at an angle so that the fluid entering the Swirler will rotate in a tangential manner.

Swirl Type Injector

In this type of atomizers, the fuel will enter the atomizer axially through the passages as shown in the figure. Later, the fuel flows into the vortex chamber which is conical in construction. As it moves along the conical part of the nozzle, a tangential velocity component is imparted to the fluid. This tangential velocity increases with the decrease in the conical radius and hence, the fluid leaves the final nozzle orifice with a high tangential velocity. In such nozzle, the fuel is made to enter the fuel nozzle tangentially so that a wider cone angle between 80-100° can be achieved. By doing this, the fuel can easily be broken down into fine droplets, thus providing a desired fuel distribution in the combustion chamber. The droplets size will further depend on the

injection pressure. Based on different injection pressures, different types of droplet pattern can be observed as shown in figure. From experiments, it is found that the droplet diameter is dependent on mass flow rate, pressure drop across the nozzle, surface tension, and viscosity.

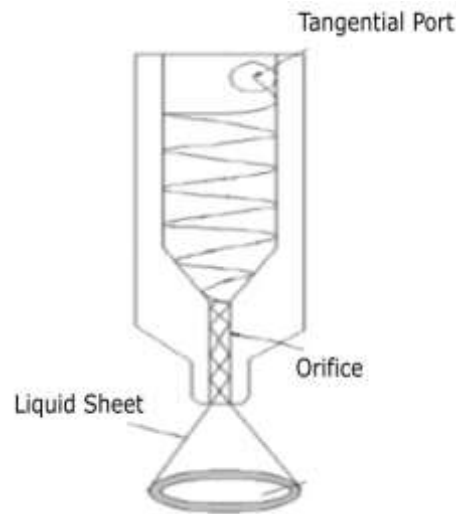


Fig: Swirl type injector used in rocket thrust chamber

Coaxial swirl injectors

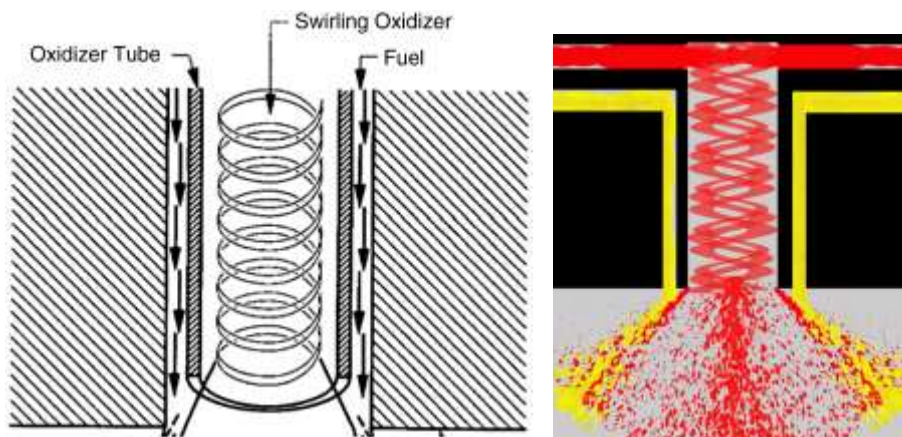


Figure- Coaxial swirler type injector

Coaxial swirl injectors are the most commonly used in present day. It will have concentric tubes through which the propellants flow. The central tube will behave much like a swirl injector produce a conical jet. But the fluid passing through the outer tube will issue out in an axial fashion as a cylindrical jet. But, in coaxial Swirler, the central jet and the outer jet collide downstream to the injector and results in proper mixing. This type of injector is commonly used for liquid oxidiser and gaseous hydrogen combination (In case of pre-burners or gas generators, gas propellant is used). It works well when the liquid hydrogen has absorbed heat from cooling jackets and has been gasified. This gasified hydrogen flows at high speed (typically 330 m/sec or 1000 ft/sec); the liquid oxygen flows far more slowly (usually at less than 33 m/sec or 100 ft/sec) and the differential velocity causes a shear action, which helps to break up the oxygen stream into small droplets. Here the speed at which the propellants are supplied through the injector is different. The differential speed of the flow of each propellant will try to pull the other jet. Due to this a shearing action between the two streams of jet moving at different speeds will break apart resulting in fine droplets. The location of the spray is also important. The propellants should not be spray farther so that its hits the chamber walls. Doing this will damage the wall during combustion and result in poor mixing, localised hot spots, and any other transient phenomenon.